

The Impact of Floods on Well-Being: Evidence from Germany, 2021

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The 2021 German Flood

- In July 2021, extreme heavy rains caused one of the deadliest flood disasters in modern European history.
- Germany was the hardest-hit country, with severe destruction in western regions, around 190 fatalities, and major damage to infrastructure and housing.
- Climate change is increasing the frequency and severity of extreme weather events such as floods. (e.g. 2024 flood)



Flood damage in Germany, July 2021

Motivation

- Floods generate substantial economic and social costs beyond immediate physical damages.
- Some consequences are immediate and visible, others may persist long after the disaster itself.
- **Direct effects:**
 - Mortality and injuries
 - Property and infrastructure damages
 - Income and employment losses
- **Indirect effects:**
 - Psychological distress and anxiety
 - Financial worries and uncertainty
 - Disruptions to social networks and daily life
 - Changes in climate risk perceptions



2021 Flood Damage

Previous Literature and Research Gap

- Early studies show that floods significantly reduce subjective wellbeing and generate substantial welfare losses Lamond et al. (2015); Luechinger and Raschky (2009); Sekulova and Van den Bergh (2016); Van Ootegem and Verhofstadt (2016)
- Other research show that climate shocks can have persistent effects on wellbeing Barua and Banerjee (2020); Twiddy et al. (2022); von Möllendorff and Hirschfeld (2016)
- Other studies show that climate shocks can affect: beliefs, risk perceptions, Labor market outcomes, earnings, and employment Demski et al. (2017); Eskander and Fankhauser (2022); Osberghaus and Fugger (2022); Parida and Chowdhury (2021); Rentschler et al. (2022)
- However, much less is known about why wellbeing losses persist over time and which mechanisms drive this persistence.

Did the July 2021 floods in Germany lead to persistent declines in subjective well-being, and what mechanisms explain these long-term effects?

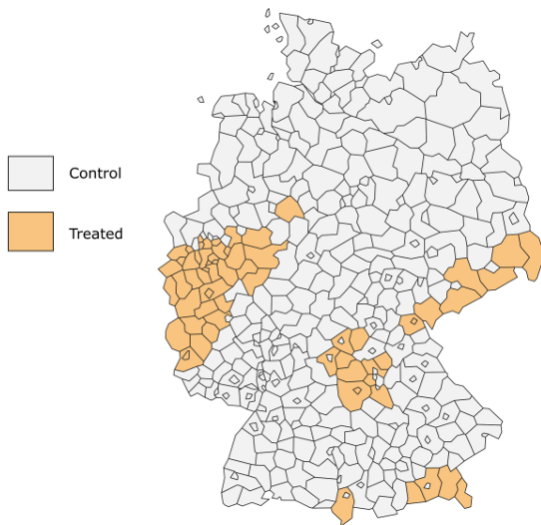
- Understanding these longer-term effects is important for designing recovery and adaptation policies.
- The flood provides a natural experiment to study the persistent effects of climate shocks on subjective wellbeing.

Three main source of Data

- German Socio-Economic Panel (SOEP)
 - SOEP contains rich information on socioeconomic characteristics, health, attitudes, and subjective well-being
 - SOEP track individuals and households in Germany annually since 1984.
- Copernicus Emergency Management Service (CEMS)
 - High-resolution geospatial shapefiles
 - District-level flood exposure from EMSR517 and EMSN105 shapfiles
- The German Federal Office for Civil Protection and Disaster Assistance (BBK)

Data: German Districts Affected by the 2021 Floods

- Germany has 400 districts, and 66 were affected by the 2021 flood.
- Individuals matched with flood-affected districts in the German floods of 2021
- Map: German Districts Affected by the 2021 Floods



- Four self-reported measures of subjective well-being from the German Socio-Economic Panel (SOEP):
 - **Life Satisfaction** (0–10) 0 = completely dissatisfied, 10 = completely satisfied
 - **Health Satisfaction** (0–10)
 - **Sleep Satisfaction** (0–10)
 - **Self-Rated Health Status** (5-point scale: bad to very good)
- Higher values indicate better well-being outcomes.

Identification Strategy

Empirical Strategy

- Difference-in-Differences (DiD) framework
- Individual and year fixed effects

Treatment Definition

- Treated individuals: residents of districts affected by the July 2021 floods
- Individuals who moved across districts are excluded to ensure consistent treatment exposure
- Respondents interviewed after July 2021 are excluded to avoid contamination from immediate flood effects

Final Sample

- Main analysis focuses on 2021–2023
- 10,538 observations
- 2,002 treated and 8,536 control observations

$$Y_{i,t} = \alpha_i + \delta_t + \sum_{k=1}^2 \beta_k (Treated_i \times \mathbb{1}[t - 2021 = k]) + v_{i,t} \quad (1)$$

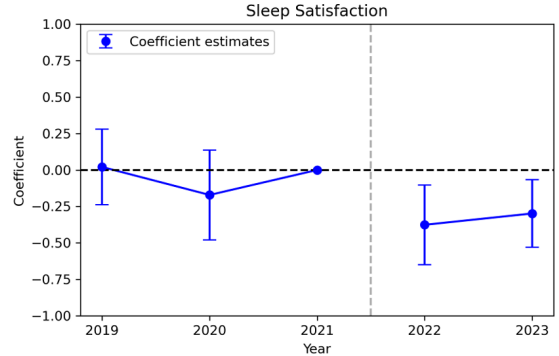
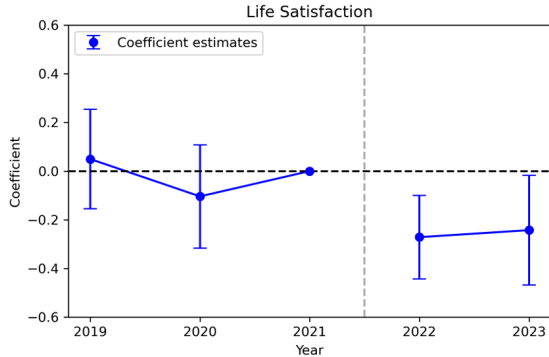
- $\mathbb{1}[t - 2021 = k]$ is an indicator for event time relative to the 2021 flood.
- $k = 1$ corresponds to the year 2022, $k = 2$ corresponds to the year 2023
- The coefficients β_k measure the treatment effects in each post-flood year relative to 2021.
- $Treated_i$ is equal to one for individuals residing in flood-affected districts
- Individual fixed effects α_i and year fixed effects δ_t
- Standard errors are clustered at the district level (NUTS-3) to account for within district correlation.

Table: Persistent Effects of the 2021 German Floods on Subjective Well-Being

	(1) Life Satisfaction	(2) Health Satisfaction	(3) Current Health	(4) Sleep Satisfaction
Treated $\times$ (year = 2022)	-0.119** (0.051)	-0.026 (0.033)	-0.028 (0.070)	-0.149* (0.089)
Treated $\times$ (year = 2023)	-0.102* (0.058)	-0.020 (0.029)	-0.097 (0.067)	-0.058 (0.090)
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Observations	12,108	12,108	12,108	5,619

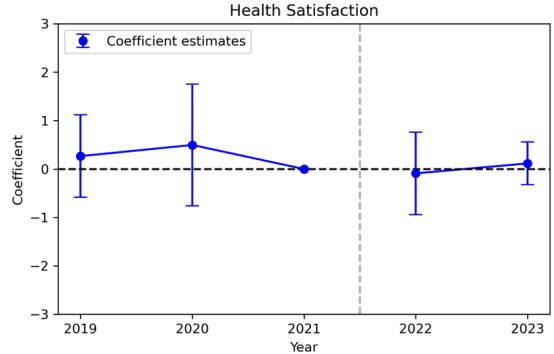
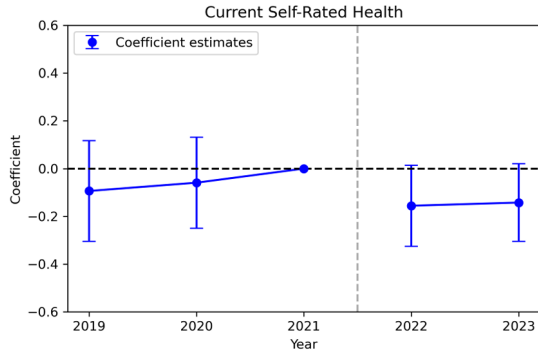
Notes: Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Pre-Trends and Robustness



Notes: The figures present event-study estimates and robustness checks for the impact of the July 2021 German floods on subjective well-being outcomes. The absence of significant pre-treatment effects supports the parallel trends assumption.

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Why Do Well-Being Effects Persist Over Time?

Income Channel

Table: Income Effects of the 2021 German Floods

	(1) Wage	(2) Employment	(3) Monthly HH Income
Treated $\times$ (year = 2022)	460.178 (618.626)	0.013 (0.013)	72.944 (80.009)
Treated $\times$ (year = 2023)	555.267 (866.827)	0.003 (0.014)	-50.711 (116.003)
Year FE	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Observations	12,108	12,108	9,909

Why Do Well-Being Effects Persist Over Time?

Climate-Related Risk Perceptions

Table: Effects of the 2021 Flood on Environmental Risk Perceptions

	(1) Environmental protection worries	(2) Climate Change worries
Treated $\times$ (year = 2022)	0.101*** (0.034)	0.075** (0.033)
Treated $\times$ (year = 2023)	0.100*** (0.025)	0.052 (0.035)
Year FE	Yes	Yes
Individual FE	Yes	Yes
Observations	7,335	7,335

Notes: Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Why Do Well-Being Effects Persist Over Time?

Housing Damage and Financial Stress

Table: Effects of the 2021 Flood on Housing Costs and Financial Stress

	(1) Maintenance Cost Last year	(2) Own Finance Worries	(3) Dwelling Satisfaction
Treated $\times$ (year = 2022)	939.090 (1343.946)	0.061 (0.042)	-0.027 (0.081)
Treated $\times$ (year = 2023)	4057.341** (1864.957)	0.071* (0.041)	-0.114 (0.081)
Year FE	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Observations	2,430	7,335	7,335

Notes: Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Why Do Well-Being Effects Persist Over Time?

Macroeconomic and Social Concerns

Table: Effects of the 2021 Flood on Economic and Social Worries

	(1) Economic Worries	(2) Job Security Worries	(3) Social Cohesion Worries
Treated $\times$ (year = 2022)	0.045 (0.040)	0.006 (0.028)	0.007 (0.037)
Treated $\times$ (year = 2023)	-0.009 (0.041)	0.004 (0.025)	0.003 (0.041)
Year FE	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Observations	7,335	7,335	7,335

Notes: Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Why Do Well-Being Effects Persist Over Time?

Health Concerns and Healthcare Utilization

Table: Effects of the 2021 Flood on Health Concerns and Healthcare Utilization

	(1)	(2)
	Own Health Worries	Annual Doctor Visits
Treated $\times$ (year = 2022)	0.050 (0.031)	0.318 (0.664)
Treated $\times$ (year = 2023)	0.012 (0.035)	1.003 (0.950)
Year FE	Yes	Yes
Individual FE	Yes	Yes
Observations	7,335	11,013

Notes: Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Evidence suggests several factors contributing to the decline:

- Changes in income **do not account for the observed decline in well-being**
- Individuals show **increased concern about environmental risks**
- Individuals show **increased worries about climate change**
- Floods cause **housing damages and repair costs** ($\approx 4,000$) and **heighten concerns about individuals' financial situation**
- No evidence on social worries, job security worries or healthcare utilization concerns.

Conclusion and Policy Implications

- The 2021 German flood caused persistent declines in subjective wellbeing.
- The results suggest that these effects are not primarily driven by income losses, but rather by:
 - Increased climate and environmental concerns
 - Financial insecurity related to housing damages and repair costs
- Climate shocks therefore generate important non-material and psychological costs that may persist long after the disaster itself.
- Policies aimed at disaster recovery should consider not only physical reconstruction, but also financial security, risk communication, and long-term resilience.

Thank you

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